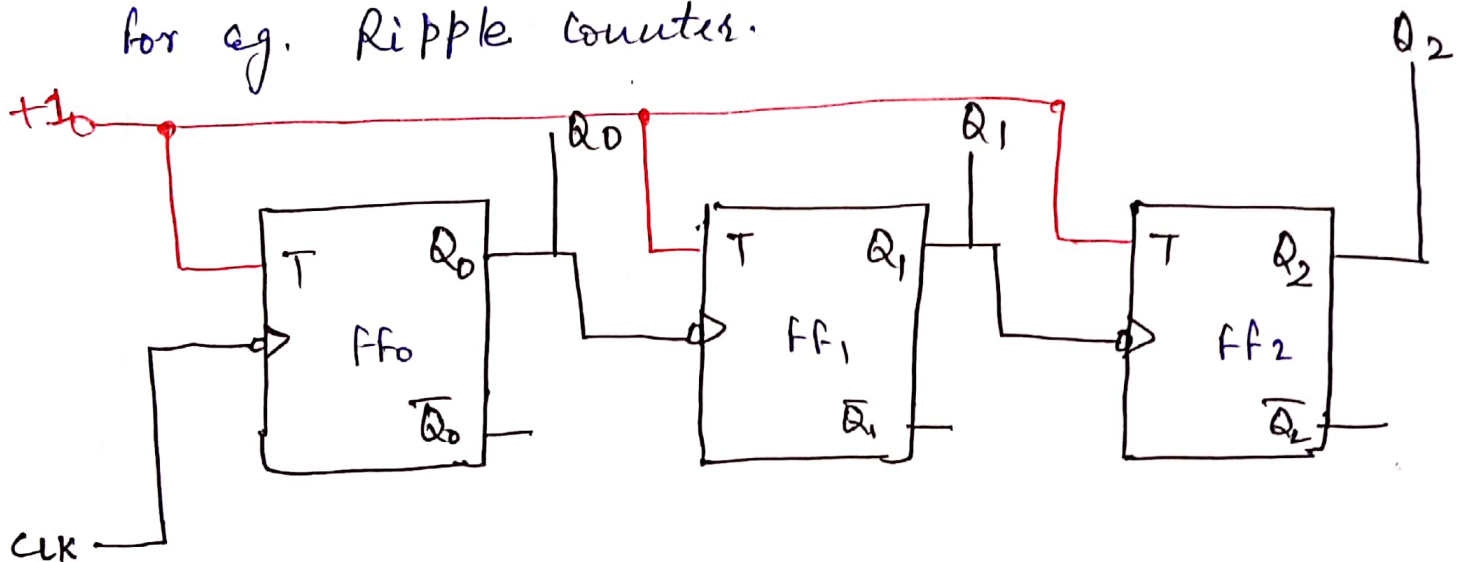


Asynchronous Counter $\Rightarrow$

(10)

- * Asynchronous Counter refers to state that does not have a fixed time relationship with each other. (FF)
- * In this type of counter flip-flops does not have a common clock pulse. so their states does not change exactly at same time.
- * In asynchronous counter 1st flip-flop has CLK and the o/p of each flip-flop acts as a CLK to the next flip-flop.

for eg. Ripple counter.

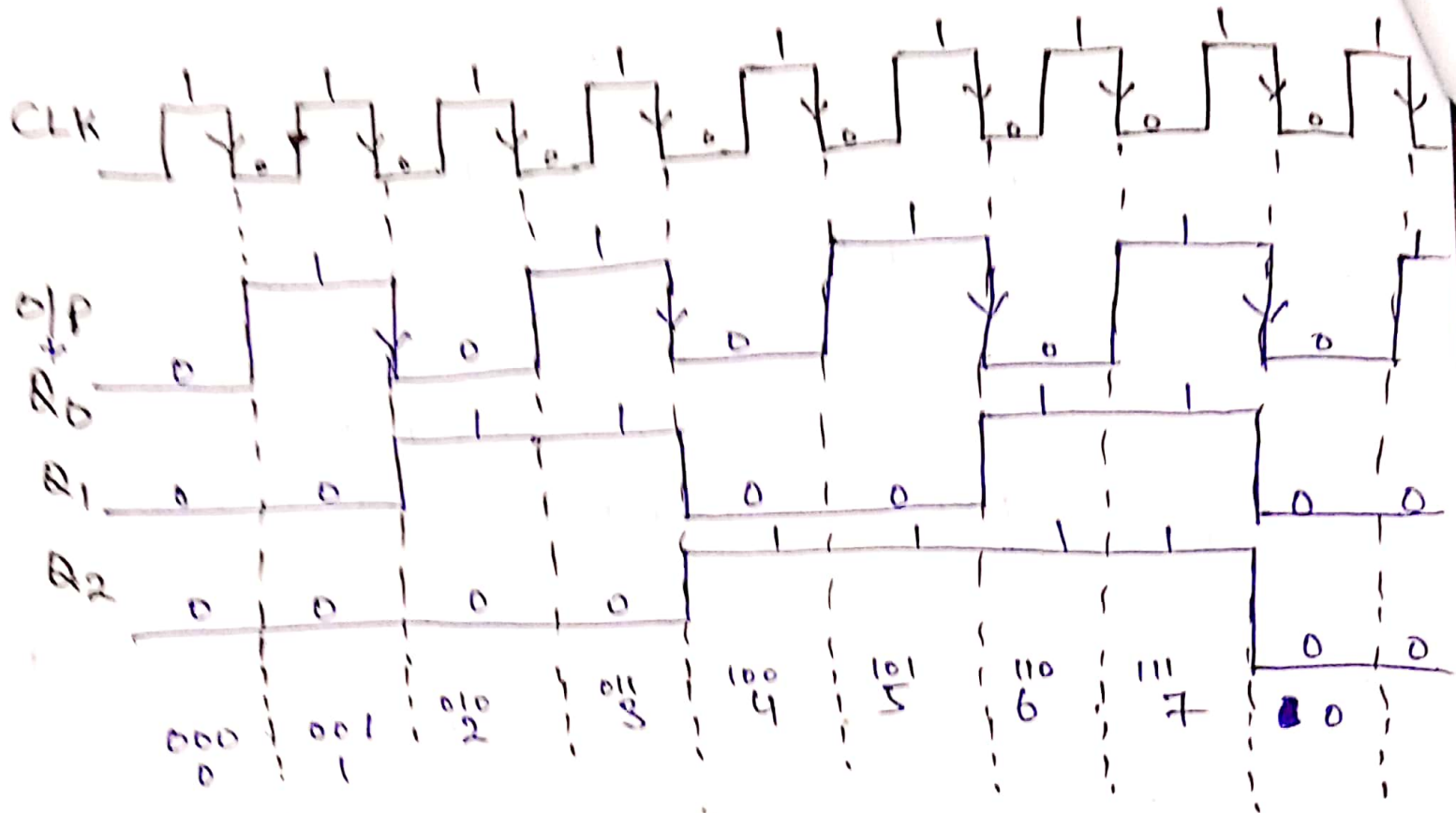


- * In this we can use T or J-K FF.

$T = J, K = 1$ is kept for all FFs.

- * First FF is connected to the external clock and second FF's CLK is connected to the o/p of 1st FF. This process is followed for the rest of the FFs.

Consider all the ffs are -ve edge triggered,
the o/p is calculated as follows.



i.e. Q_0 toggle at -ve edge of the clock.

Q_1 " " " " " " Q_0 i.e. (ff0)
 Q_2 " " " " " " Q_1 i.e. (ff1)

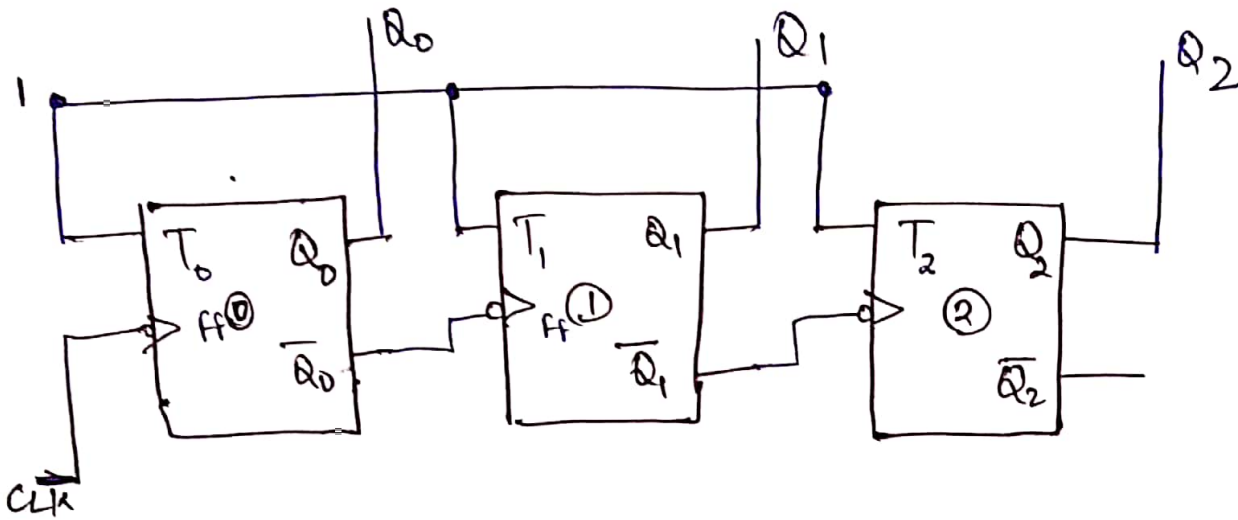
* This way circuit gives o/p from $Q_2 Q_1 Q_0 = 000, 001, 010, 011, 100, 101, 110, 111$.

* This work as Mod-8 ripple counter (UP count)

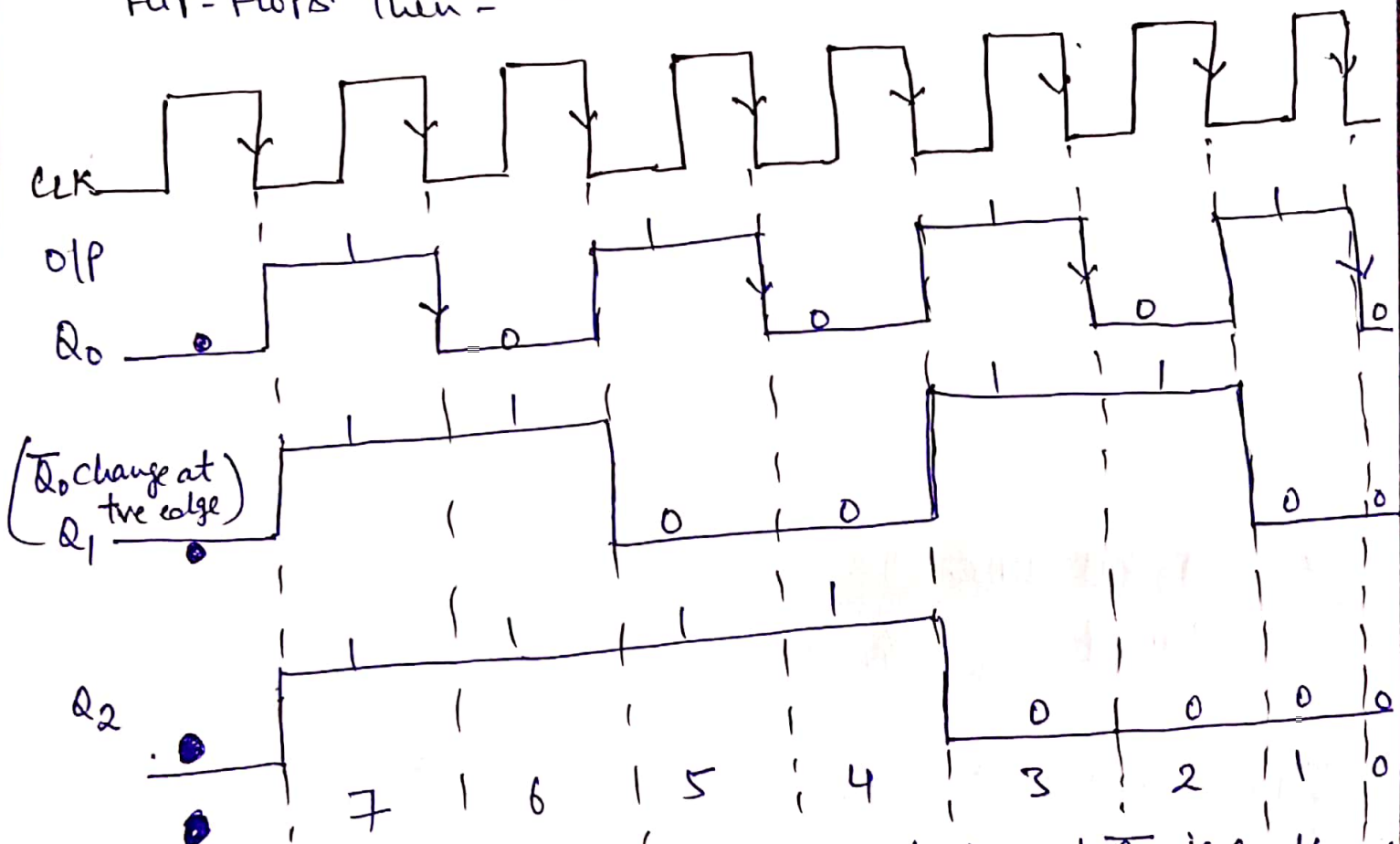
Down Counter $\rightarrow$

* In the down counter the complement of the O/P of 1st flip-flop will act as a CLK of the next ff.

* Rest is same as in up counter.



In the above diagram these are -ve edge triggered flip-flops. Then -



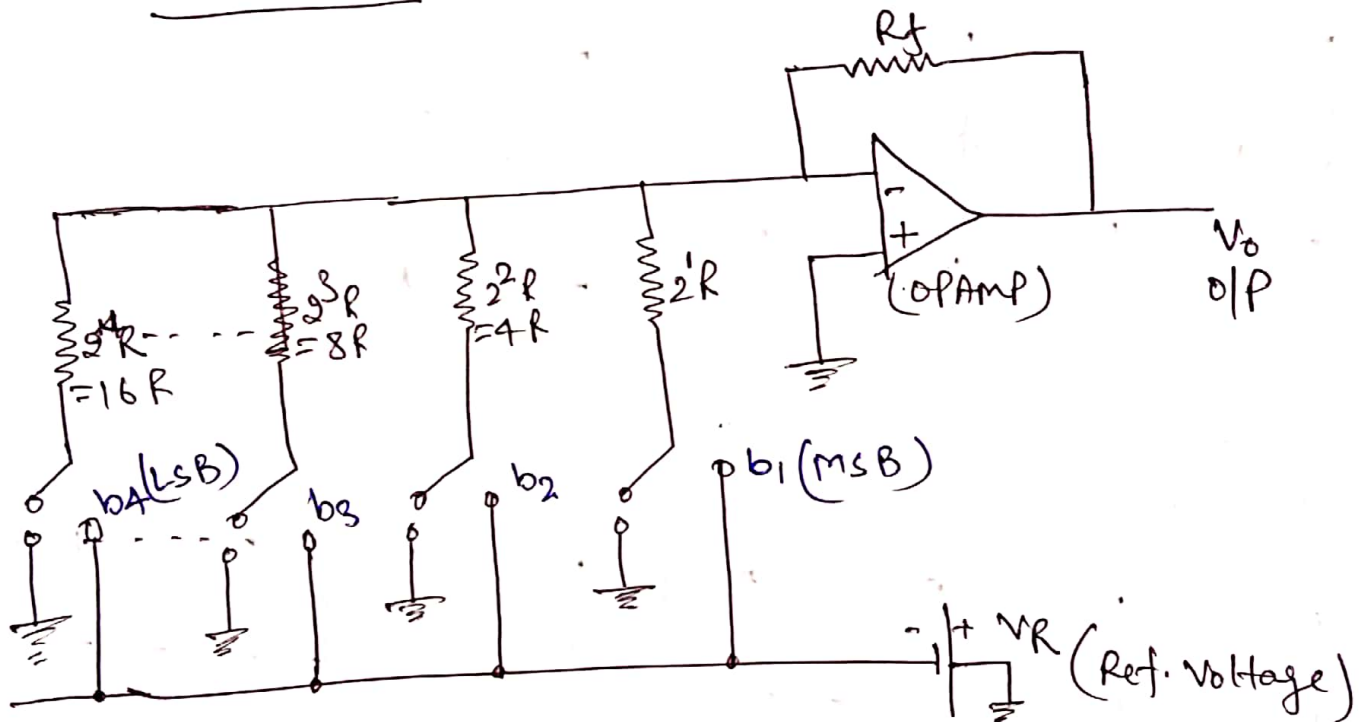
Here Q_0 toggle with -ve edge of the clock and $\bar{Q}_0$ is complement of Q_0 . Q_1 toggle with the edge of Q_0 because $\bar{Q}_0$ is complement of Q_0 . Similarly, Q_2 toggle with the edge of Q_1 because $\bar{Q}_1$ is complement of Q_1 .

Digital to Analog Converters:-

A D/A Converter (DAC) converts digital data into its equivalent analog data.

→ There are different types of D/A Converters-

① Binary weighted Resistor (DAC) :-



(For ~~4~~ bit data converter)
(It may up to N-bit)

4-bit

* In a D/A Converter, which accepts the data from 0000 to 1111. i.e. there are 15 discrete levels of i/p's and

* In 4-bit DAC the 2^0 bit is changed to $\frac{1}{15^{\text{th}}}$ of V_R , 2^1 bit is changed $\frac{2}{15^{\text{th}}}$ of V_R similarly 2^3 bit to $\frac{8}{15^{\text{th}}}$ of V_R .

For example if the data is ^{3bit} 1010.

for 101

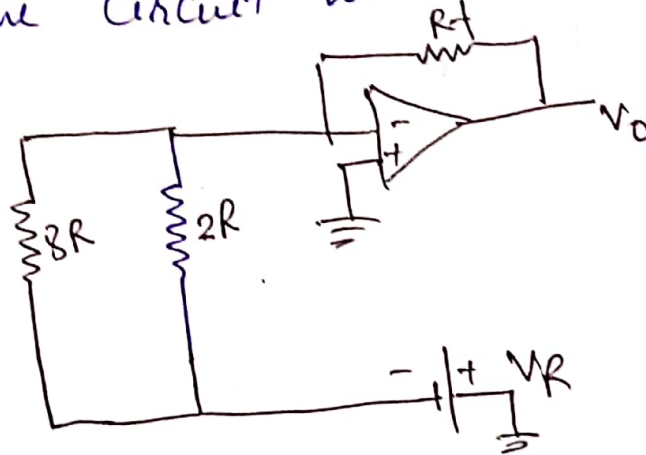
(12)

$$b_1 = 1, b_2 = 0, b_3 = 1, \text{ ~~10001~~}$$

when the i/p value is 1 then respective switch will be close and if i/p value is 0 then that switch will behave as a open switch.

i.e. for 101

the circuit will be



(10)

$$2R \parallel 8R$$

$$\text{i.e. } \frac{1}{2} + \frac{1}{8} = \frac{16}{10} R$$

$$\text{Now the } V_o = - \frac{R_f}{R_{in}} \times V_R$$

$$= - \frac{R_f}{\frac{16}{10} R} \times V_R$$

$$\text{if } R_f = R$$

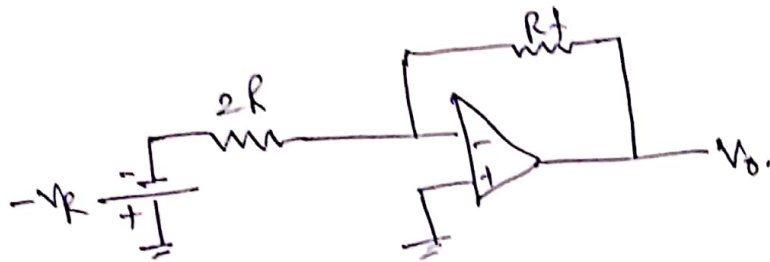
$$\text{then } V_o = \frac{R}{\frac{16}{10} R} V_R = \frac{10}{16} V_R$$

$$\boxed{V_o = \frac{10}{16} V_R}$$

for data 001 i.e. ①

i.e. $b_1=1, b_2=0, b_3=0$

then circuit will be



$$\text{i.e. } V_o = -\frac{R_f}{2R} \times V_R$$

$$\text{i.e. } V_o = -\frac{R_f}{2R} \times V_R$$

$$\text{if } R_f = R$$

$$\text{then } V_o = \frac{R}{2R} \times V_R$$

$$V_o = \frac{1}{2} V_R$$

Similarly for from 000 to 111 we can calculate the value of V_o (output voltage)

then a general formula for V_o is

$$V_o = \left[\frac{b_1}{2^1} + \frac{b_2}{2^2} + \frac{b_3}{2^3} + \dots + \frac{b_N}{2^N} \right] \times V_R$$

for 101 i.e. 5

$$V_o = \left[\frac{1}{2} + \frac{1}{8} \right] \times V_R$$

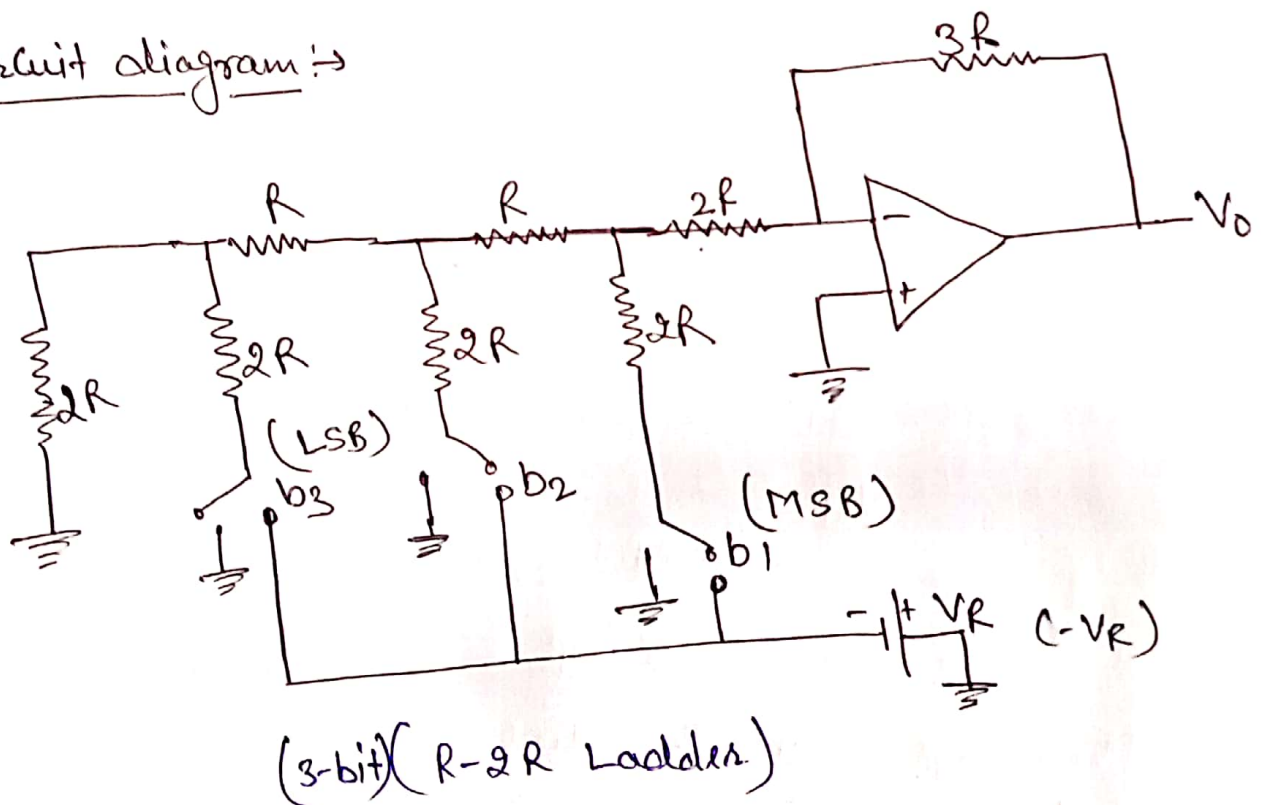
$$\text{i.e. } \left(V_o = \frac{10}{16} V_R \right) \text{ as previous.}$$

② R-2R Ladder (DAC) $\Rightarrow$ (Voltage Switches)

(13)

- * In weighted resistor and R-2R ladder types of D/A Converters the current flowing through the resistor changes as the i/p data changes.
- * The main advantages of R-2R type D/A converters are
 - * More accurate selection and design of Resistor R and 2R are possible.
 - * The binary word length can be increased by adding required no. of R-2R sections.

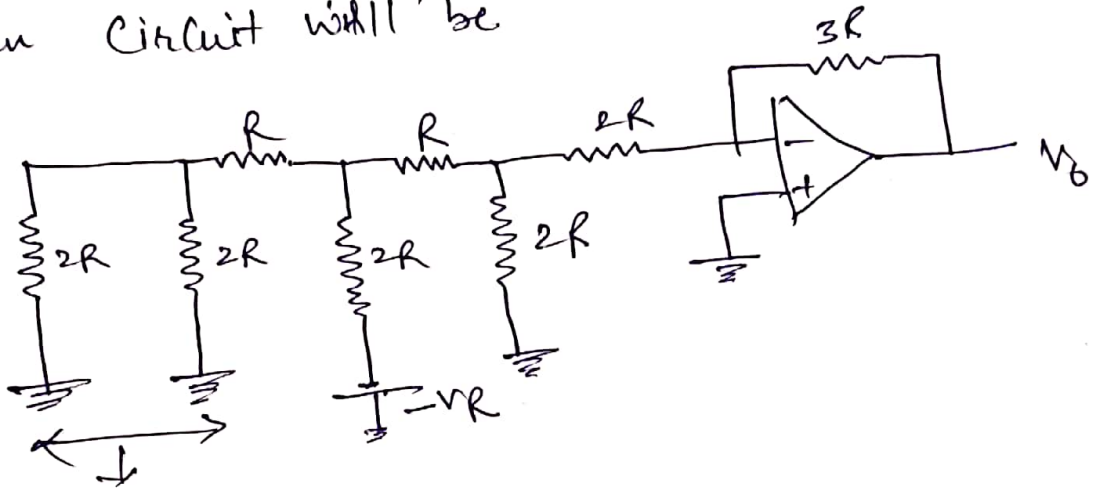
Circuit diagram $\Rightarrow$



In this type of Converter if we want to increase the no. of bit at the i/p end then No Complexity increase as in binary weighted.

For example the DIP is 010. (3 bit data)
ie. $b_1 = 0, b_2 = 1, b_3 = 0$

then circuit will be



$$2R \parallel 2R$$

then total resistance will R .

Next R will be in series so $R + R = 2R$

Next $2R$ and $2R$ will be in parallel

then total will be R with series R

then $2R$.

Now $2R$ and $2R$ with $(-VR)$ will be parallel.

we have to apply Thevenin theorem to calculate V_{th}

$$V_{th} = \frac{-VR \times 2R}{2R + 2R} = \frac{-VR}{2}$$

and $R_{th} = 2R \parallel 2R = R$

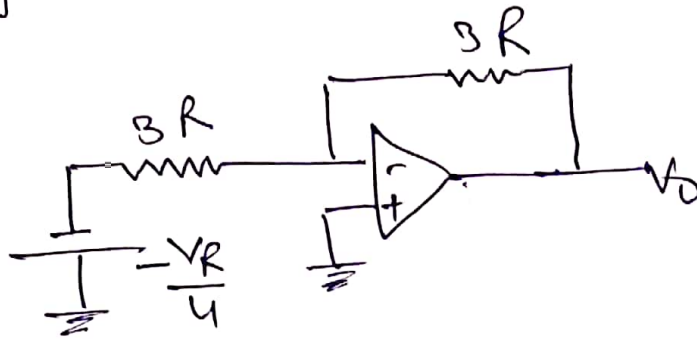
then $2R$ with potential $-\frac{VR}{2}$ and $2R$ with ground. so we have to use Thevenin theorem again

$$V_{th} = \frac{\frac{-VR}{2} \times 2R}{2R + 2R} = \frac{-VR}{4}$$

and $R_{th} = 2R \parallel 2R = R$

Finally circuit will be

(14)



then

$$V_O = -\frac{R_f}{R_{in}} \times V_{in}$$

then

$$V_O = -\frac{3R}{3R} \times -\frac{V_R}{4}$$

i.e. $\boxed{V_O = -\frac{V_R}{4}}$ for data 010.

Similarly we can apply the data from 000 to 111. and general formula for R-2R Ladder type DAC is

$$\boxed{V_O = \left[\frac{b_N}{2^N} + \frac{b_{N-1}}{2^{N-1}} + \dots + \frac{b_2}{2^2} + \frac{b_1}{2^1} \right] V_R}$$

Analog to Digital (ADC or A/D) Converter $\Rightarrow$

- * An A/D Converter does the inverse of the D/A. It converts analog signal into its equivalent n -bit binary code digital o/p signal.
- * The digital o/p from an A/D Converter can be in Serial or Parallel form.

① Simultaneous type or Flash type A/D Converter $\Rightarrow$

The Simultaneous type A/D Converter is based on comparing an unknown analog i/p voltage with a set of reference voltages.

- * To Convert ~~analog~~ an analog signal into a digital signal of N -bit output bits $(2^N - 1)$ No. of comparators are required.

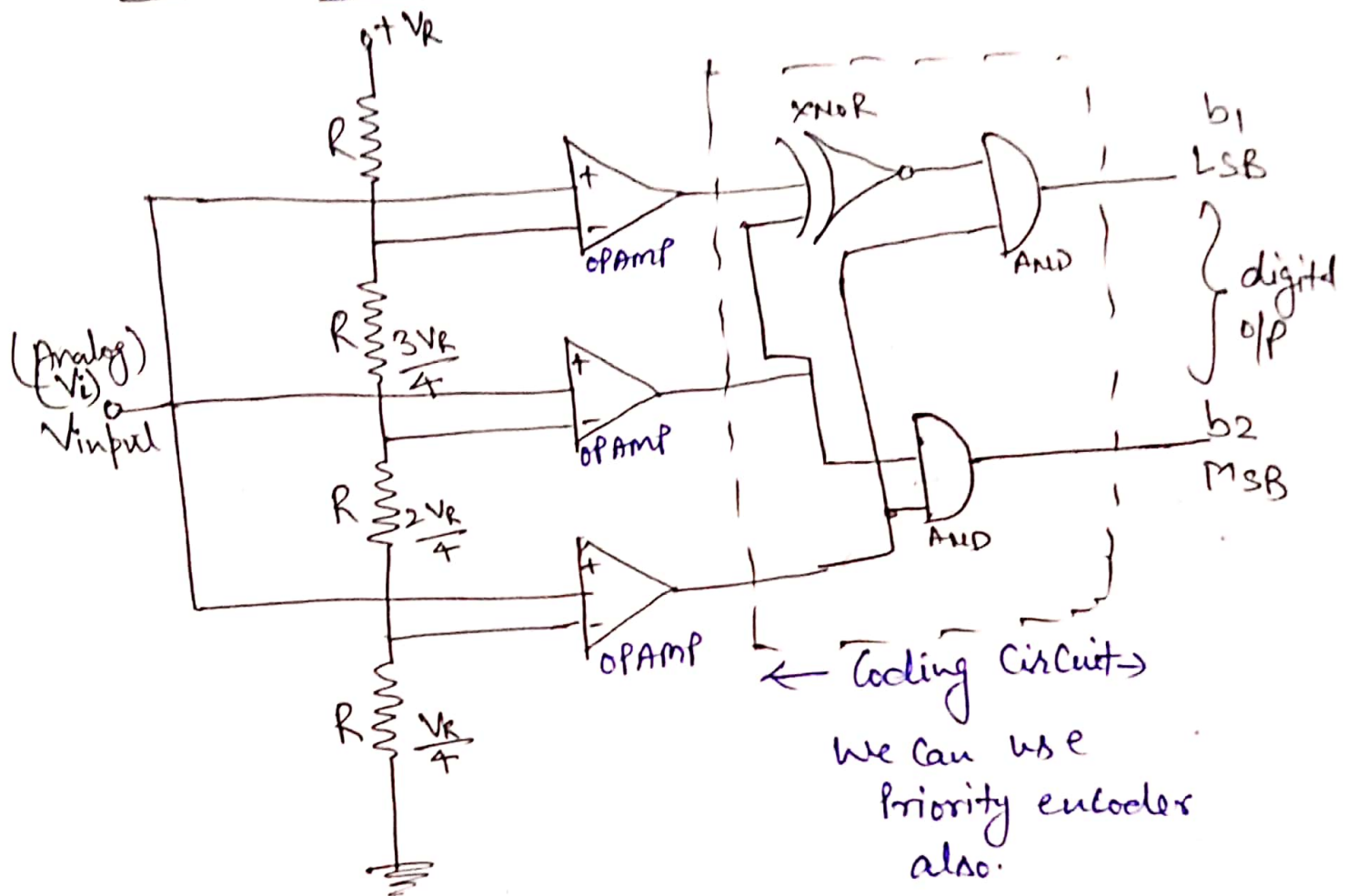
for example 2-bit A/D Converter requires $(2^2 - 1) = 3$ comparators.

while a 3-bit A/D Converter requires $2^3 - 1 = 7$ comparators.

- * No. of resistors are 2^N for N -bit digital o/p.
i.e. for 2-bit Converter total 4 resistors are required.

2-bit A/D Converter -

(15)



* In the above circuit there are three Comparators (OPAMP) and 4 Resistor.

The voltage drop across all the resistors ~~are~~ is fixed i.e. from the ground side resistor having $V_r/4$, then $2V_r/4$, $3V_r/4$ (for ~~2~~ 2-bit) - if for 3 bit data total 8 resistor will require and 7 comparators.

the drop will be $V_r/8$, $2V_r/8$, $3V_r/8$ up to $7V_r/8$.

Now if $V_{in} > V_{ref}$ then the o/p of the OPAMP will be 1 (high).

if $V_{in} < V_{ref}$ then o/p of the OPAMP = 0 (low)

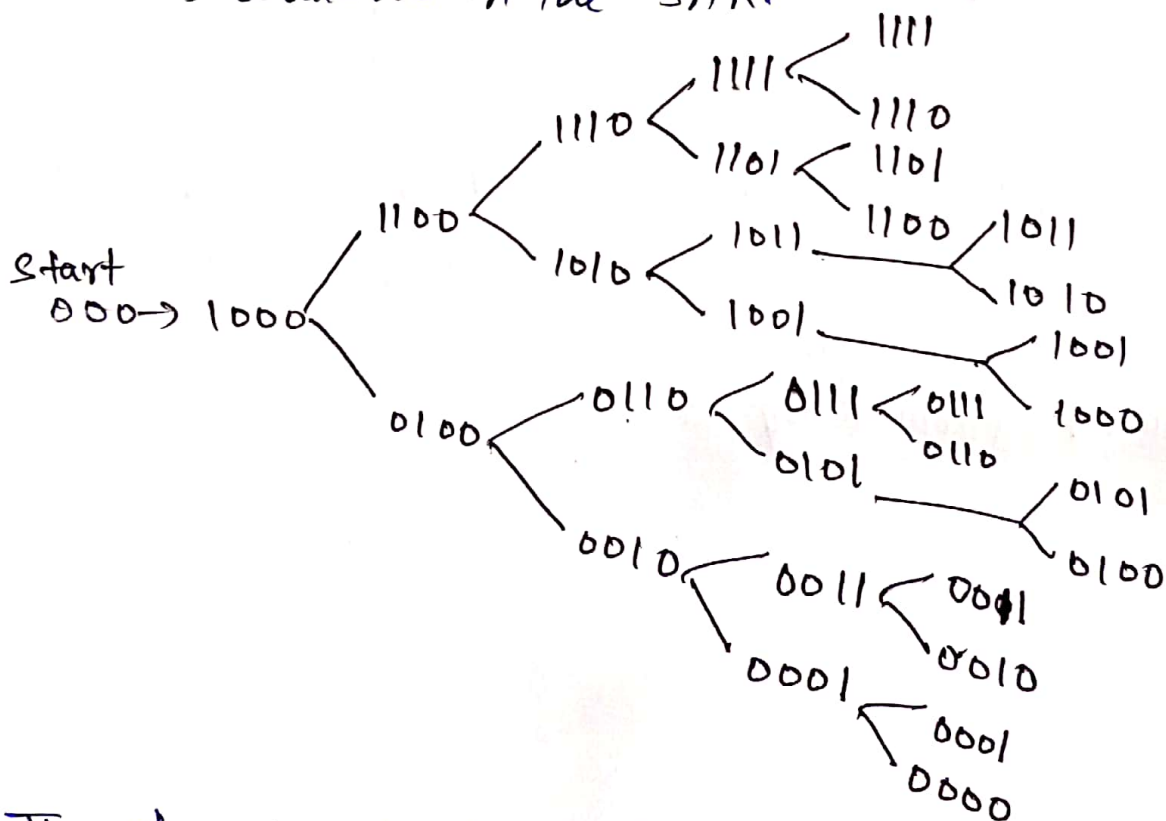
Using Comparator Principle we can generate the digital o/p. that o/p will give to the encoding circuit or to a Priority encoder and that particular analog value can be converted into digital signal.

Analog I/P	Comparator o/p			digital o/p	
	c_1	c_2	c_3	b_2	b_1
$0 \leq V_i \leq V/4$	0	0	0	0	0
$V/4 \leq V_i \leq V/2$	1	0	0	0	1
$V/2 \leq V_i \leq 3V/4$	1	1	0	1	0
$3V/4 \leq V_i \leq V$	1	1	1	1	1

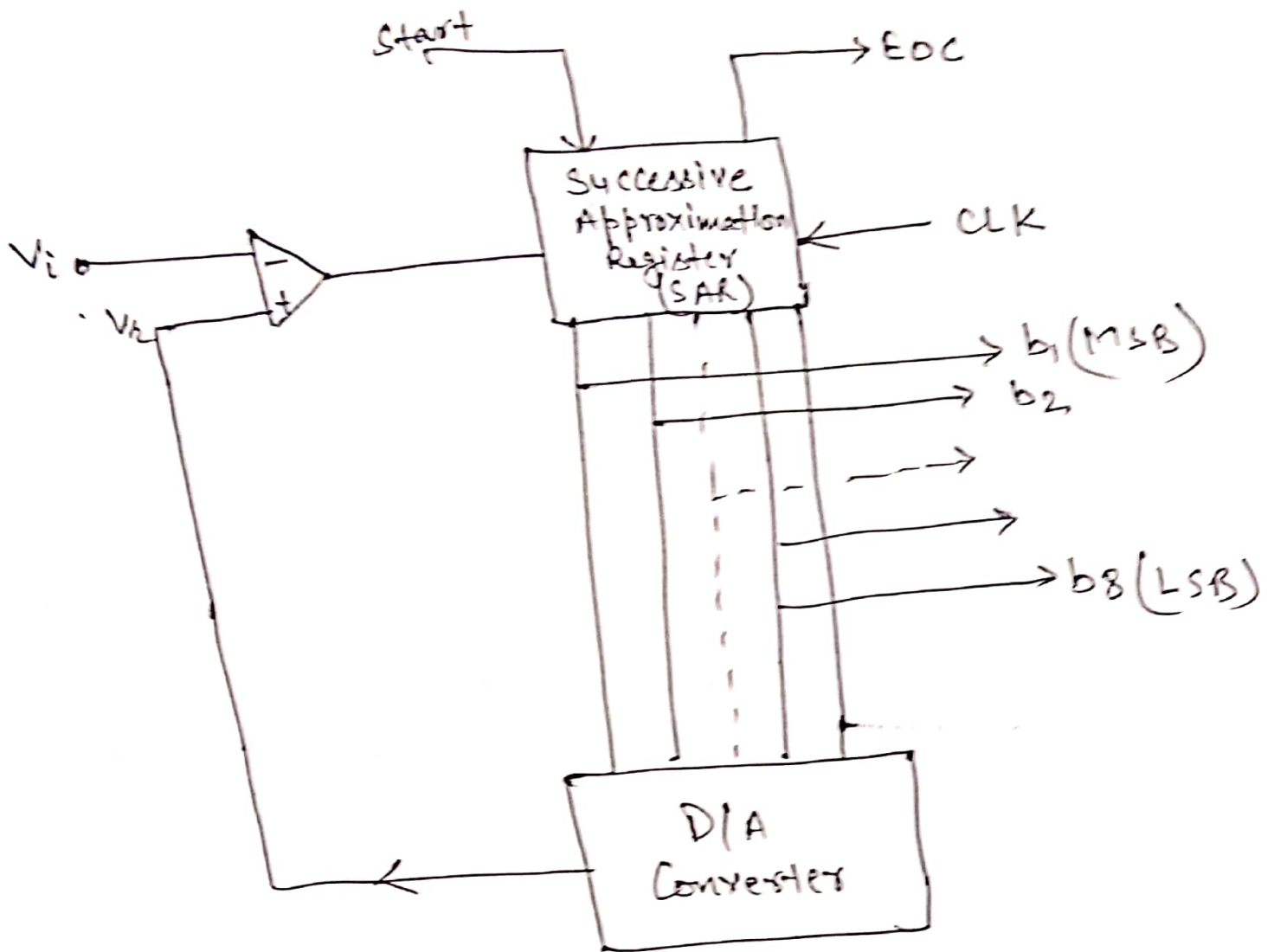
② Successive approximation type A/D Converter $\rightarrow$

- * The Conversion time is maintained constant in successive approximation type A/D Converter, and it is proportional to the number of bits in the digital o/p, unlike the counter and continuous type A/D Converters.
- * This type of A/D Converter operates by successively dividing the voltage range by half as explained in the following steps —

- ① The MSB is initially set to 1 with the remaining three bits set as 0. The digital equivalent is compared with the unknown analog i/p voltage.
- ② If the analog voltage is higher than the digital equivalent, the MSB is retained as 1 and the second MSB is set to 1. otherwise the MSB is reset to 0 and the second MSB is set to 1.
- ③ Comparison is made as given in step 1 to decide whether to retain or reset the second MSB. The third MSB is set to 1 and the operation is repeated down to LSB and by this time the converted digital value is available in the SAR.

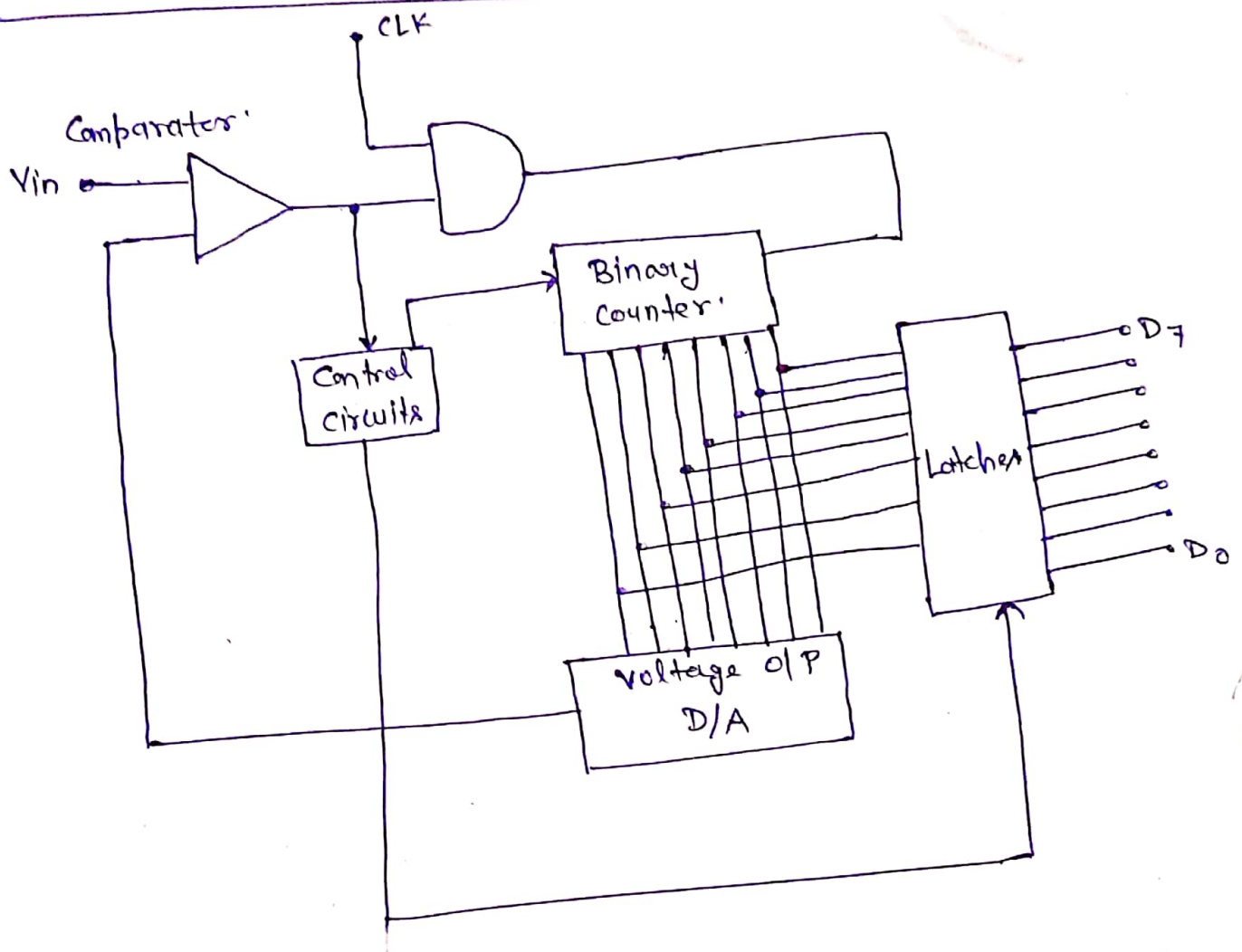


The functional block diagram of successive approximation type A/D Converter is as shown in figure.



When Start Command is applied, the SAR sets the MSB (b_1) of the digital signal, while the other bits are made zero, so that the trial code becomes 1 followed by zeros.

Counter type A/D Converter →



- * In this system, a Continuous sequence of pulses is passed through a gate. At the start of the conversion cycle, the counters are reset to zero, so the o/p of the D/A is at 0V.
- * A positive unknown voltage is applied to the i/p of Converter will cause the o/p of the Comparator to go high and enable the AND gate. This will let the clock pulses to the Counter. Each clock pulse increments the counter by 1 and increases the voltage on the output of the D/A Converter by one step.
- * When the voltage on the o/p of D/A Converter passes the voltage on the unknown V voltage, the o/p of the Comparator will go low and shut off the clock pulses to the Counter. The count accumulated on the Counter is proportional to the i/p.

voltage. The Control circuitry then strobes the latches to transfer the count to the o/p and resets the counters to start another conversion cycle.